

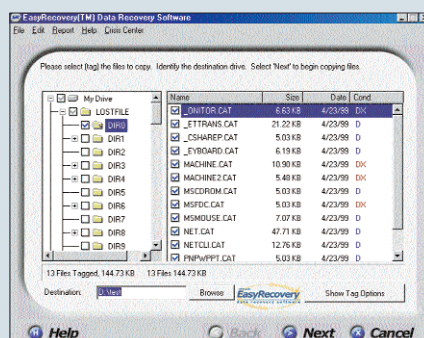
On the road to recovery

Need to recover your data from a crashed hard disk? Well, don't be disappointed. Here's how you go about doing it. (In this example, we have taken Ontrack Easy Recovery, and Lost & Found.)

■ As soon as you realise your hard disk has crashed, shut down your PC, so as to prevent any further damage. Then restart it, go to the BIOS, and check if the hard disk is being detected. If your hard disk is not detected, you won't be able to recover data through any software. You will have to send it to the data recovery centre.

■ Next, connect another hard disk to your PC to store the recovered data. However, ensure that the free disk space is more than the size of the data you need to restore; else the task won't be successful. Alternatively, you can plug in a zip drive, but make sure that the recovery software you're using supports such media. Note that you cannot recover data onto another partition that is present on the crashed hard disk. Hence, plugging in an additional storage media on which the data can be stored becomes mandatory.

■ Next, make sure you have a bootable floppy disk. Insert the floppy disk, and boot your



Recover lost files using Ontrack Easy Recovery

machine in the MS-DOS mode. Go into the BIOS, and change the boot sequence from there so as to make the PC boot from the floppy disk.

■ After you've booted in DOS mode, insert the recovery software floppy diskette. Here we've taken the example of Ontrack's Easy Recovery software, which is considered to be one of the most widely used, and effective recovery software. You can also use other software such as Stellar Phoenix Data Recovery Software, Final Data, etc.

■ After inserting the recovery software, you get an option to select the primary hard disk

(the one that crashed) as the recovery source, and the second hard disk as the restore destination.

■ The software then analyses the data that it has to recover from the crashed hard disk. In order to recover the data, the software tries to read the two copies of the FAT tables, and then reads the raw data on the hard drive, trying to match it to the FAT Tables. The data that matches the FAT table is the data that is visible to you, even after the crash. The data that does not match the FAT table is generally the data that you are trying to recover. Once it comes across raw data on the disk that corresponds to a file, but finds that there is no similar entry in the FAT table, it marks the data as a recoverable file, and adds it to the existing list of files it has from the FAT table. All the files that exist on the hard drive, whether in the FAT table, or detected by the software, are shown as recoverable files. Once this is done, you can choose the data to be recovered, which will be copied to the destination drive. The whole process might take hours to complete, depending purely on how much data has to be recovered.

important to note down the CPU utilisation of the drive, since a high CPU utilisation, would slow down the entire system.

In the HD-Tach benchmark, nearly all drives performed on par with each other, with the read burst speed hitting the theoretical limit of 80 MBps, indicating that all the drives have good on-board controllers. The only drive that could not make it to the 80 MBps mark was the Samsung SV0602H, which registered a speed of 78 MBps—not a bad score, but comparatively slower than the rest. An important point to note was that all the drives were able to hit the 80 MBps mark, but at the cost of higher CPU utilisation.

In the SATA category, the Maxtor DiamondMax Plus9 160 GB drive sported the highest CPU utilisation of around 4.1 per cent. This was the only drive in the entire comparison test that exceeded 3 per cent CPU utilisation. With



Seagate Barracuda ST3160023AS

such a high CPU utilisation, it is quite evident that the performance of this drive will be hampered in the real world environment that often involves multi-tasking. For example, if you are working on a document, and copying some files in the background simultaneously, the higher CPU utilisation will result in more CPU time being diverted to the copying of files rather than the document that you're working on.

The Seagate 160 GB ST3160023AS drive took second last place by utilising 2.1 per cent of CPU time. Both these drives had an unusually high CPU utilisation, as compared to the other drives in the comparison test. The 200 GB Maxtor DiamondMax Plus9 drive took the crown with

the least amount of CPU utilisation, by tipping the pointer at just 0.4 per cent. Its lower capacity sibling—the 120 GB DiamondMax Plus 9, followed it.

Among the IDE drives, the Samsung's

SV0411N fared badly by utilising 2.5 per cent of CPU time. This was followed by Maxtor MaxLine Plus II, and Samsung's SP0802N that utilised 1 per cent of the CPU time during the test. Seagate's ST340015A and Samsung's SV1203N, took the top honours, utilising just 0.6 per cent of the CPU time.

SiSoft Sandra tests

The SiSoft Sandra tests came up with

